# Abelian Improvement Proposal 0015: Revisit: Mnemonic Codes for Generating Deterministic Accounts *

Abelian

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**Abstract**. This AIP is a revisited version of AIP0011. To be self-completed, this AIP adds a standalone section based on the AIP0011. In the added section, this AIP describes how an additional root seed, say CoinVKRootSeedAut, as a part of Account-Root-Seeds, is derived from the Entropy-Seed.

This CoinVKRootSeedAut is added to separate the root seed for value-hidden key for CTAUT from that for full-private Abelian coin, to provide more flexible extension in the future.

To support 12/15/18/21-word mnemonics, a section is added to include the mapping algorithms between mnemonics and entropy-seed. In addition, some notes on PRF and EntropySeedToMasterSeed algorithms are included in the added section.

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# Contents


**1 Entropy-Seeds and Mnemonics Conversion** **4**

1.1 Sample An Entropy-Seed . . . . . 4

1.2 From Entropy-Seeds To Mnemonics . . . . . 4

1.3 From Mnemonics To Entropy-Seeds . . . . . 5

1.4 Test vectors . . . . . 6

**2 Derivation of Account-Root-Seeds from Entropy-Seeds** **6**

2.1 Preliminaries on Abelian Wallet Account . . . . . 6

2.2 Derivation From Entropy-Seeds to Master-Seeds . . . . . 8

2.3 Derivation From Master-Seed to Account-Root-Seeds . . . . . 9

2.4 Deterministic Public Rands . . . . . 10

2.5 Test Vectors . . . . . 11

**3 Compatibility** **11**

3.1 A Previous Version . . . . . 11

3.2 Compatibility . . . . . 12

**4 Revisit: Derivation From Master-Seed to Account-Root-Seeds** **13**

4.1 Revisit: Derivation From Master-Seed to Account-Root-Seeds . . . 13

4.2 Compatibility . . . . . 15

**5 Revisit: Supporting 12-21 word Mnemonics** **15**

5.1 Sample An Entropy-Seed . . . . . 15

5.2 From Entropy-Seeds To Mnemonics . . . . . 16

5.3 From Mnemonics To Entropy-Seeds . . . . . 17

5.4 Test vectors . . . . . 19

5.5 Derivation From Entropy-Seeds to Master-Seeds . . . . . 19

**A WordList** **21**

**B Test vectors For Mapping Between Mnemonics and Entropy-Seeds** **27**

**C Revisit: Test vectors For Mapping Between Mnemonics and Entropy-Seeds** **29**

**D Test vectors For Derivation of Master-Seeds and Account-Root-Seeds** **36**

# 1 Entropy-Seeds and Mnemonics Conversion

## 1.1 Sample An Entropy-Seed

This proposal uses an entropy of 256 bits as the random bit-string seed for a deterministic wallet account, referred to as **Entropy-Seed**.

Note that it is required that **Entropy-Seed** is sampled randomly and uniformly from $\{0,1\}^{256}$, as shown in Algorithm 1.

**Algorithm 1** $SampleEntropySeed()$

1: $entropyseed \xleftarrow{\$} \{0,1\}^{256}$
2: **return** $entropyseed$

## 1.2 From Entropy-Seeds To Mnemonics

For a given **Entropy-Seed** $entropyseed$, the corresponding mnemonic is obtained by the following Algorithm 2.

**Algorithm 2** $EntropySeedToMnemonic(entropyseed, wordlist)$

1: $cs \leftarrow the\ first\ 8\ bits\ of\ \mathsf{SHA256}(entropyseed)$
2: $ext \leftarrow entropyseed \| cs$
3: $ms_0 \| ms_1 \| \ldots \| ms_{23} \leftarrow ext$
4: **for** $t = 0$ to 23 step 1 **do**
5: $\quad i_t \leftarrow \mathsf{BinaryToInt11}(ms_t)$
6: $\quad mnemonic[t] \leftarrow wordlist[i_t]$
7: **end for**
8: **return** $mnemonic$

*Remark:*

- $entropyseed$ is an **Entropy-Seed** sampled as in Section 1.1.
- Here SHA256 is described in the standard of SHA2 [4].
- $ext$ consists of 256+8 = 264 bits.

- $ext$ is split into 24 groups of bits, say $ms_0, ms_1, \ldots, ms_{23}$, each consisting of 11 bits, such that $ms_0 \| ms_1 \| \ldots \| ms_{23} = ext$.
- $\mathsf{BinaryToInt11}()$ is the **standard** algorithm that converts binary-string in $\{0,1\}^{11}$ to the corresponding decimal integer in $[0, 2047]$. In particular, $\mathsf{BinaryToInt11}(000, 0000, 0001) = 1$, $\mathsf{BinaryToInt11}(000, 0000, 1000) = 8$, $\mathsf{BinaryToInt11}(100, 0000, 0000) = 1024$, and so on.
- $wordlist$ is the commonly used wordlist with 2048 words as in BIP0039 [5], as shown in Appendix A.
- As a result, the output $mnemonic$ consists of 24 words in $wordlist$.

## 1.3 From Mnemonics To Entropy-Seeds

The mapping from Mnemonics to Entropy-Seeds is just the inverse procedure as shown in the following Algorithm 3.

**Algorithm 3** $MnemonicToEntropySeed(mnemonic, wordlist)$

**for** $t = 0$ to 23 step 1 **do**
  $i_t \leftarrow \mathsf{LookupIndex}(mnemonic[t], wordlist)$
  **if** $i_t \notin [0, 2047]$ **then**
    **return** FAIL
  **end if**
  $ms_t \leftarrow \mathsf{IntToBinary11}(i_t)$
**end for**
$ext \leftarrow ms_0 \| ms_1 \| \ldots \| ms_{23}$
$entropyseed \leftarrow the\ first\ 256\ bits\ of\ ext$
$cs \leftarrow the\ last\ 8\ bits\ of\ ext$
$cs' \leftarrow the\ first\ 8\ bits\ of\ \mathsf{SHA256}(entropyseed)$
**if** $cs' \neq cs$ **then**
  **return** FAIL
**end if**
**return** $entropyseed$

*Remark:*

- *mnemonic* consists of 24 words.
- *wordlist* is the commonly used wordlist with 2048 words as in BIP0039, as shown in Appendix A.
- LookupIndex$(word, wordlist)$ finds the index of *word* in *wordlist*. Note that if the output index is not in the scope $[0, 2047]$, it implies that an illegal word is used and FAIL is returned.
- IntToBinary11() is the inverse of BinaryToInt11(), converting an integer in $[0, 2047]$ to a binary-string in $\{0,1\}^{11}$.
- The output *entropyseed* is a 256-bit string in $\{0,1\}^{256}$ .

### 1.4 Test vectors

The test vectors for the mapping between Mnemonics and Entropy-Seeds are given in Appendix B.

## 2 Derivation of Account-Root-Seeds from Entropy-Seeds

### 2.1 Preliminaries on Abelian Wallet Account

**Account and Account-Root-Seeds.** In Abelian, as shown in Fig. 1, each **account** consists of a set of **root seeds/keys**, referred to as **Account-Root-Seeds**, say (CoinSpKeyRootSeed, CoinSnKeyRootSeed, CoinDetectorRootKey, CoinVKeyRootSeed), where CoinSnKeyRootSeed and CoinVKeyRootSeed are optional. In particular, for an account which will generate only pseudo-private addresses, the CoinSnKeyRootSeed

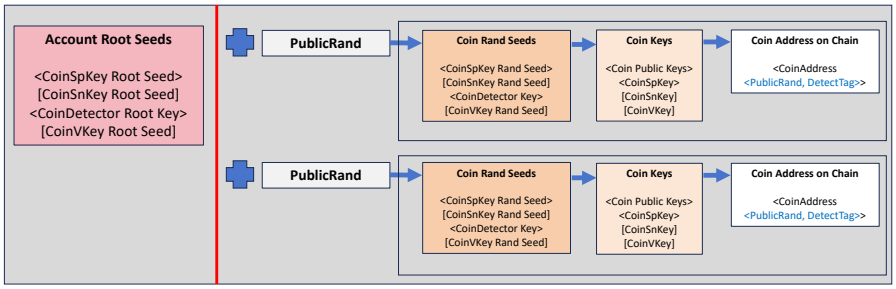



Figure 1: Account and (Address, key)

and CoinVKeyRootSeed are null. Note that CoinSpKeyRootSeed, CoinSnKeyRootSeed, CoinDetectorRootKey, and CoinVKeyRootSeed are all 512-bit long.

**Public Rand and (Address, Key).** As shown in Fig. 1, to generate an (Address, Key) pair under an account, a **Public Rand** (with 512 bits) needs to be introduced. In particular, for each given Public Rand, a corresponding (Address, Key) will be *deterministically* generated from the Account-Root-Seeds. Note that there are two ways to generate an (address, key) pair under an account, namely (1) given only the Account-Root-Seeds: sample a random Public Rand and generate the (address, key) pair from the given Account-Root-Seeds and sampled Public Rand, or (2) given the Account-Root-Seeds and a well-form Public Rand: generate the (address, key) pair from the given Account-Root-Seeds and Public Rand.

**Derivation of Account-Root-Seeds.** For such an account, it is ideal that CoinSpKeyRootSeed, CoinSnKeyRootSeed, CoinDetectorRootKey, and CoinVKeyRootSeed are independent entropies. However, from the view of practice, it is desired that they are derived from an entropy, referred to as Master-Seed, as shown in Fig. 2. As a response to such a desire, to provide good user-friendliness (say, using mnemonic), this proposal derives a Master-Seed from Entropy-Seed, and then derives Account-Root-Seeds from the Master-Seed.

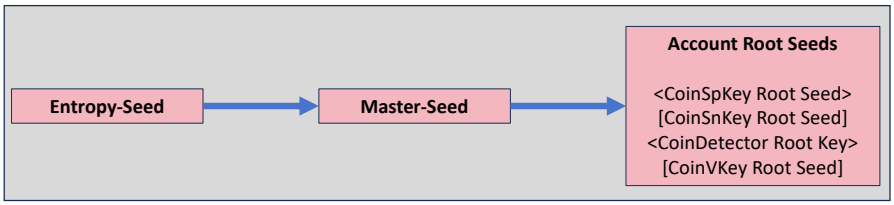



Figure 2: Derivation of Account-Root-Seeds

Note that this proposal does not directly use Entropy-Seed as Master-Seed. This is to provide flexibility to potential extension, for example, deriving multiple accounts from one Entropy-Seed.

**Notations.** Below we define a PRF (Pseudo Random Function)

$$\mathsf{PRF}(key, input) := \mathsf{KMAC256}(key, input, 512, \text{“ABELIANPRF”})$$

where KMAC256 serves as a PRF [2] (approved by NIST) to generate 512-bit output, *input* serves as the context, and “ABELIANPRF” specifies the Domain Separation Customization String. Note that here with a 512-bit (or 256-bit) key and a 512-bit output, the above PRF provides 256-bit security.[1]

## 2.2 Derivation From Entropy-Seeds to Master-Seeds

Given an Entropy-Seed, a corresponding Master-Seed is deterministically derived as shown in the following Algorithm 4. Note that with the security of PRF, it can be deduced that given a master-seed, it is infeasible to distinguish the Entropy-Seed from a random $x \in \{0,1\}^{256}$.

[1]Why use KMAC256 with a 256-bit input and a 512-bit output rather than HMAC-SHA512? To be safe, we want to use SHA3 rather than SHA2 (i.e., SHA512), and KMAC is the only known (variable-length message) authentication code algorithm based on KECCAK which is the underlying function of the SHA3 standard. Although it is claimed that HMAC can be based on any Hash function, we are not sure whether the claim is applicable to SHA3, since HMAC was

**Algorithm 4** $EntropySeedToMasterSeed(entropyseed, customizationContext)$

1: $masterseed \leftarrow$ PRF($entropyseed$, "AccountMasterSeed"||customizationContext)

2: **return** $masterseed$

*Remark:*

- *entropyseed* is 256-bit long, determined by the 24-word mnemonic rule.
- *entropyseed* serves as the key.
- "AccountMasterSeed"||customizationContext serves as the input, where different applications may use different customizationContext (which is "" by default). This also allows to support the case of generating multiple accounts from one mnemonic.
- *masterseed* is 512-bit long.

Note that this derivation is very different from that of BIP0039.

## 2.3 Derivation From Master-Seed to Account-Root-Seeds

Given a Master-Seed, the corresponding Account-Root-Seeds are deterministically derived as shown in the following Algorithm 5.

**Algorithm 5** $MasterSeedToAccountRootSeeds(masterseed)$

1: coinSpKeyRootSeed $\leftarrow$ PRF($masterseed$, "CoinSpendKeyRootSeed")
2: coinSnKeyRootSeed $\leftarrow$ PRF($masterseed$, "CoinSerialNumberKeyRootSeed")
3: coinDetectorRootKey $\leftarrow$ PRF($masterseed$, "CoinDetectorRootKey")
4: coinVKeyRootSeed $\leftarrow$ PRF($masterseed$, "CoinValueKeyRootSeed")
5: **return** (coinSpKeyRootSeed, coinSnKeyRootSeed, coinDetectorRootKey, coinVKeyRootSeed)

proposed before SHA3.

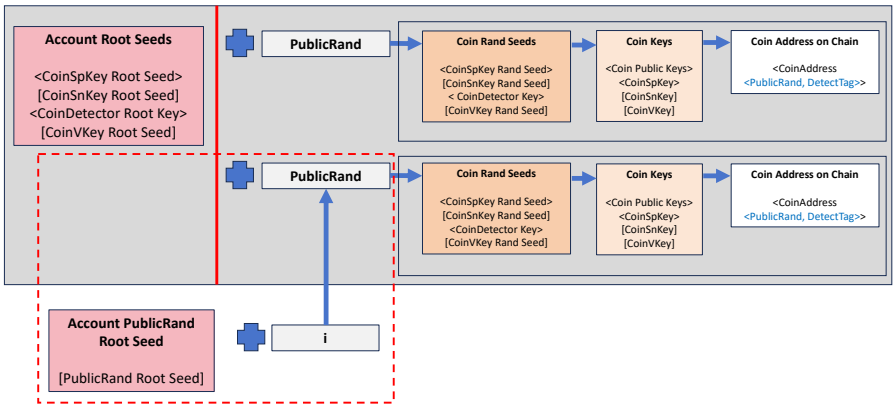



Figure 3: Account with Deterministic Public Rands

## 2.4 Deterministic Public Rands

In some scenarioes, it is desired that deterministic Public Rands are used. To support such scenarioes, this proposal defines a rule to deterministically generate Public Rands from integers in $[0, 2^{32} - 1]$, referred to as "Sequence Numbers". In particular, this proposal derives a PublicRandRootSeed for an account from a given Master-Seed, and then derives Public Rands from given sequence numbers when needed, as shown in Fig. 3, Algorithm 6, and Algorithm 7.

**Algorithm 6** $MasterSeedToAccountPublicRandRootSeed(masterseed)$

1: publicRandRootSeed ← PRF($masterseed$, "PublicRandRootSeed")
2: **return** publicRandRootSeed

**Algorithm 7** $DerivePublicRand(publicRandRootSeed, i)$

1: $seqNo$ ← EncodeSeqNo($i$)
2: publicRand ← PRF($publicRandRootSeed, seqNo$)
3: **return** publicRand

*Remark:*

- publicRandRootSeed output by $MasterSeedToAccountPublicRandRootSeed()$ is 512-bit long.

- $i$ is an integer in $[0, 2^{32} - 1]$.
- $\mathsf{EncodeSeqNo}(i)$ encodes $i \in [0, 2^{32} - 1]$ to a hex-string of length 8 with lower case, say $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, a, b, c, d, e, f$. In particular, $\mathsf{EncodeSeqNo}(1) = 00000001$, $\mathsf{EncodeSeqNo}(10) = 0000000a$, $\mathsf{EncodeSeqNo}(31) = 0000001f$.
- The output $\mathsf{publicRand}$ is 512-bit long.

### 2.5 Test Vectors

The test vectors for the derivation from Entropy-Seeds to Master-Seeds and the derivation from Master-Seeds to Account-Root-Seeds are given in Appendix D.

## 3 Compatibility

### 3.1 A Previous Version

At $height = 300,000$, we released packages of $\mathsf{AbewalletMLP\text{-}v1.0.1}$, which follow BIP0039, namely

- The mapping between Entropy-Seeds and Mnemonics is the same as Algorithm 2 and Algorithm 3.
- The derivation from Entropy-Seeds to Master-Seeds is shown as the following Algorithm 8.
- The derivation from Master-Seeds to Account-Root-Seeds is shown as the following Algorithm 9.

*Remark:*

**Algorithm 8** $EntropySeedToMasterSeed_{BIP39}(entropyseed)$

1: $words \leftarrow EntropySeedToMnemonic(entropyseed)$
2: $key \leftarrow Use\ whitespace\ to\ splice\ words$
3: $masterseed \leftarrow$ PBKDF2($key$, "mnemonic", 2048, 64, HMAC-SHA512)
4: **return** $masterseed$

**Algorithm 9** $MasterSeedToAccountRootSeeds_{old}(masterseed)$

1: coinSpKeyRootSeed $\leftarrow$ PRFOLD($masterseed$, "spendkey")
2: coinSnKeyRootSeed $\leftarrow$ PRFOLD($masterseed$, "serialnumberkey")
3: coinDetectorRootKey $\leftarrow$ PRFOLD($masterseed$, "detectorkey")
4: coinVKeyRootSeed $\leftarrow$ PRFOLD($masterseed$, "valuekey")
5: **return** (coinSpKeyRootSeed, coinSnKeyRootSeed, coinDetectorRootKey, coinVKeyRootSeed)

- PBKDF2 [1] applies a pseudorandom function (such as HMAC-SHA512) repeatedly to the salt and password to generate the key. The NIST Recommendation [6] approved PBKDF2 as the PBKDF using HMAC with any approved hash function as the PRF, and decided to revise it [3].
- Here HMAC-SHA512 is used as the pseudo-random function [5].
- PRFOLD is defined by :

  PRFOLD($key, input$) := KMAC256($key, input$, 512, "PQABELIAN-WALLET")

  where KMAC256 serves as a PRF [2] to generate 512-bit output, *input* serves as the context, and "PQABELIAN-WALLET" specifies the Domain Separation Customization String.

## 3.2 Compatibility

Note that there may be only a few of users that created mnemonics by using AbewalletMLP-v1.0.1, and that a compatibility solution transparent to users may be pretty complicated and may cause huge development effort and inefficiency, we could use a simple way to address the compatibility.

In particular, at the UI-layer of a wallet, the user is noticed that if his mnemonic was generated by AbewalletMLP-v1.0.1, he should tell the system (for example by a check box) and the system will call the above Algorithm 8 and Algorithm 9 accordingly.

# 4 Revisit: Derivation From Master-Seed to Account-Root-Seeds

## 4.1 Revisit: Derivation From Master-Seed to Account-Root-Seeds

Given a Master-Seed, the corresponding Account-Root-Seeds are deterministically derived as shown in the following Algorithm 10.

**Algorithm 10** $MasterSeedToAccountRootSeeds(masterseed)$

1: coinSpKeyRootSeed $\leftarrow$ PRF($masterseed$, "CoinSpendKeyRootSeed")
2: coinSnKeyRootSeed $\leftarrow$ PRF($masterseed$, "CoinSerialNumberKeyRootSeed")
3: coinDetectorRootKey $\leftarrow$ PRF($masterseed$, "CoinDetectorRootKey")
4: coinVKeyRootSeed $\leftarrow$ PRF($masterseed$, "CoinValueKeyRootSeed")
5: coinVKeyRootSeedAut $\leftarrow$ PRF($masterseed$, "CoinValueKeyRootSeedAut")
6: **return** (coinSpKeyRootSeed, coinSnKeyRootSeed, coinDetectorRootKey, coinVKeyRootSeed, coinVKeyRootSeedAut)

Note that, compared with the original Algorithm 5, only a coinVKeyRootSeedAut is added. Fig. 4 and Fig. 5 shows the change as well. It is worth noticing that when generating an address, at most one of {coinVKeyRootSeed, coinVKeyRootSeedAut} is used. Please refer to the following Sec. 4.2 for the details.

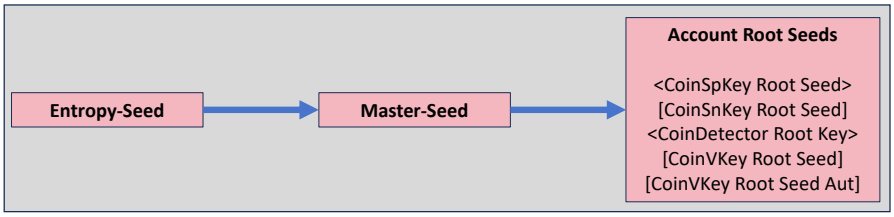



Figure 4: Revisit: Derivation of Account-Root-Seeds

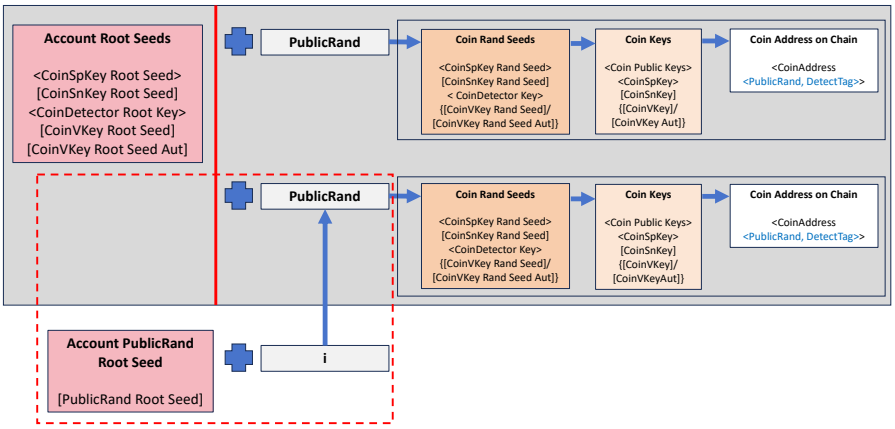



Figure 5: Revisit: Account with Deterministic Public Rands

## 4.2 Compatibility

Before the Aconcagua Fork, the chain supports three types of address corresponding to three privacy levels, say "Type 0" for "Full-Privacy-Legacy", "Type 1" for "Full-Privacy", "Type 2" for "Pseudonym-Privacy". Type 0 address is not in the scope of this document.

From the Aconcagua Fork, an additional type of address is added, say "Type 3" for "Pseudonym-CT-Privacy" where "CT" is for "Confidential Transaction" of Abelian User Token (AUT). The Type 3 address is used to replace the Type 2 address and Type 2 address is deprecated but still supported. In particular, for these addresses, the rules are as below:

- To create a Type 1 address,
  (CoinSpKeyRootSeed, CoinSnKeyRootSeed, CoinDetectorRootKey,
  CoinVKeyRootSeed) is used.

- To create Type 2 address,
  (CoinSpKeyRootSeed, CoinDetectorRootKey) is used.

- To create Type 3 address,
  (CoinSpKeyRootSeed, CoinDetectorRootKey, CoinVKeyRootSeedAut) is used.

# 5 Revisit: Supporting 12-21 word Mnemonics

## 5.1 Sample An Entropy-Seed

This revisited proposal supports an entropy of 128/160/192/224/256 bits as the random bit-string seed for a deterministic wallet account, referred to as **Entropy-Seed**.

Note that it is required that **Entropy-Seed** is sampled randomly and uniformly from $\{0,1\}^\lambda$ where $\lambda \in \{128, 160, 192, 224, 256\}$, as shown in Algorithm 11.

Note that Algorithm 11 is compatible with the Algorithm 1, where Algorithm 1 is a special case of Algorithm 11 with $\lambda = 256$. We use the same algorithm name.

**Algorithm 11** $SampleEntropySeed(\lambda)$

$entropyseed \xleftarrow{\$} \{0,1\}^\lambda$
**return** $entropyseed$

## 5.2 From Entropy-Seeds To Mnemonics

For a given **Entropy-Seed** $entropyseed$ with length $\lambda \in \{128, 160, 192, 224, 256\}$, the corresponding mnemonic is obtained by the following Algorithm 12.

Note that Algorithm 12 is compatible with the Algorithm 2, where Algorithm 2 is a special case of Algorithm 12 with $\lambda = 256$. We use the same algorithm name.

**Algorithm 12** $EntropySeedToMnemonic(entropyseed, wordlist)$

$\lambda = |entropyseed|$
**if** $\lambda \notin \{128, 160, 192, 224, 256\}$ **then**
  **return** error
**end if**
$cs \leftarrow the\ first\ \frac{\lambda}{32}\ bits\ of\ \mathsf{SHA256}(entropyseed)$
$ext \leftarrow entropyseed \| cs$
$n \leftarrow \frac{|ext|}{11}$
$ms_0 \| ms_1 \| \ldots \| ms_{n-1} \leftarrow ext$
**for** $t = 0$ to $n-1$ step 1 **do**
  $i_t \leftarrow \mathsf{BinaryToInt11}(ms_t)$
  $mnemonic[t] \leftarrow wordlist[i_t]$
**end for**
**return** $mnemonic$

*Remark:*

- $entropyseed$ is an **Entropy-Seed** sampled as in Algorithm 11.

- Here SHA256 is described in the standard of SHA2 [4].
- *ext* consists of $\lambda+\lambda/32$ bits, i.e., $132, 165, 198, 231, 264$ for $\lambda = 128, 160, 192, 224, 256$, respectively.
- *ext* is split into $n := (\lambda + \lambda/32)/11$ groups of bits, say $ms_0, ms_1, \ldots, ms_{n-1}$, each consisting of 11 bits, such that $ms_0 \| ms_1 \| \ldots \| ms_{n-1} = ext$.
- BinaryToInt11() is the **standard** algorithm that converts binary-string in $\{0,1\}^{11}$ to the corresponding decimal integer in $[0, 2047]$. In particular, BinaryToInt11$(000, 0000, 0001) = 1$, BinaryToInt11$(000, 0000, 1000) = 8$, BinaryToInt11$(100, 0000, 0000) = 1024$, and so on.
- *wordlist* is the commonly used wordlist with 2048 words as in BIP0039 [5], as shown in Appendix A.
- As a result, the output *mnemonic* consists of $n \in \{12, 15, 18, 21, 24\}$ words in *wordlist*.

## 5.3 From Mnemonics To Entropy-Seeds

The mapping from Mnemonics to Entropy-Seeds is just the inverse procedure as shown in the following Algorithm 13.

*Remark:*

- *mnemonic* consists of $n \in \{12, 15, 18, 21, 24\}$ words.
- *wordlist* is the commonly used wordlist with 2048 words as in BIP0039, as shown in Appendix A.
- LookupIndex$(word, wordlist)$ finds the index of *word* in *wordlist*. Note that if the output index is not in the scope $[0, 2047]$, it implies that an illegal word is used and FAIL is returned.

**Algorithm 13** $MnemonicToEntropySeed(mnemonic, wordlist)$

$n := |mnemonic|$
**if** $n \notin \{12, 15, 18, 21, 24\}$ **then**
  **return** error
**end if**
**for** $t = 0$ to $n - 1$ step 1 **do**
  $i_t \leftarrow \mathsf{LookupIndex}(mnemonic[t], wordlist)$
  **if** $i_t \notin [0, 2047]$ **then**
    **return** FAIL
  **end if**
  $ms_t \leftarrow \mathsf{IntToBinary11}(i_t)$
**end for**
$ext \leftarrow ms_0 \| ms_1 \| \ldots \| ms_{n-1}$
$\lambda \leftarrow \lfloor (|ext| - 1)/8 \rfloor * 8$
$entropyseed \leftarrow the\ first\ \lambda\ bits\ of\ ext$
$cs \leftarrow the\ last\ \lambda/32\ bits\ of\ ext$
$cs' \leftarrow the\ first\ \lambda/32\ bits\ of\ \mathsf{SHA256}(entropyseed)$
**if** $cs' \neq cs$ **then**
  **return** FAIL
**end if**
**return** $entropyseed$

- IntToBinary11() is the inverse of BinaryToInt11(), converting an integer in $[0, 2047]$ to a binary-string in $\{0,1\}^{11}$.
- The output *entropyseed* is a $\lambda$-bit string in $\{0,1\}^{\lambda}$ where $\lambda \in \{128, 160, 192, 224, 256\}$ depending on the number of words in the input mnemonics.

## 5.4 Test vectors

The test vectors for the mapping between Mnemonics and Entropy-Seeds are given in Appendix C.

## 5.5 Derivation From Entropy-Seeds to Master-Seeds

Given an Entropy-Seed, a corresponding Master-Seed is deterministically derived as shown in the following Algorithm 14.

Note that this algorithm is the same as Algorithm 4, also based on the underlying PRF algorithm defined in Sec. 2.1. In particular,

$$\mathsf{PRF}(key, input) := \mathsf{KMAC256}(key, input, 512, \text{“ABELIANPRF”})$$

where KMAC256 serves as a PRF [2] (approved by NIST) to generate 512-bit output, *input* serves as the context, and “ABELIANPRF” specifies the Domain Separation Customization String.

**In Sec. 2.1, it is required that the *key* is 256 bits and the PRF provides 256-bit security. Now, it is allowed that *key* could be 128/160/192/224/256 bits, and the PRF provides 128/160/192/224/256-bit security. We would like to emphasize, in Abelian, all other uses of the PRF, except the EntropySeedToMasterSeed here, still requires the key is 256 bits.**

*Remark:*

**Algorithm 14** *EntropySeedToMasterSeed*(*entropyseed*, *customizationContext*)

1: *masterseed* ← PRF(*entropyseed*, "AccountMasterSeed"||customizationContext)

2: **return** *masterseed*

- *entropyseed* is 128/160/192/224/256-bit long, determined by the 12/15/18/21/24-word mnemonic rule.
- *entropyseed* serves as the key.
- "AccountMasterSeed"||customizationContext serves as the input, where different applications may use different customizationContext (which is "" by default). This also allows to support the case of generating multiple accounts from one mnemonic.
- *masterseed* is 512-bit long.

Note that this derivation is very different from that of BIP0039.

# References


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# A WordList

The wordlist is the same as that of BIP0039 at https://github.com/bitcoin/bips/blob/master/bip-0039/english.txt. In particular, all words in sequence as following:

abandon ability able about above absent absorb abstract absurd abuse access accident account accuse achieve acid acoustic acquire across act action actor actress actual adapt add addict address adjust admit adult advance advice aerobic affair afford afraid again age agent agree ahead aim air airport aisle alarm album alcohol alert alien all alley allow almost alone alpha already also alter always amateur amazing among amount amused analyst anchor ancient anger angle angry animal ankle announce annual another answer antenna antique anxiety any apart apology appear apple approve april arch arctic area arena argue arm armed armor army around arrange arrest arrive arrow art artefact artist artwork ask aspect assault asset assist assume asthma athlete atom attack attend attitude attract auction audit august aunt author auto autumn average avocado avoid awake aware away awesome awful awkward axis baby bachelor bacon badge bag balance balcony ball bamboo banana banner bar barely bargain barrel base basic basket battle beach bean beauty because become beef before begin behave behind believe below belt bench benefit best betray better between beyond bicycle bid bike bind biology bird birth bitter black blade blame blanket blast bleak bless blind blood blossom blouse

blue blur blush board boat body boil bomb bone bonus book boost border boring borrow boss bottom bounce box boy bracket brain brand brass brave bread breeze brick bridge brief bright bring brisk broccoli broken bronze broom brother brown brush bubble buddy budget buffalo build bulb bulk bullet bundle bunker burden burger burst bus business busy butter buyer buzz cabbage cabin cable cactus cage cake call calm camera camp can canal cancel candy cannon canoe canvas canyon capable capital captain car carbon card cargo carpet carry cart case cash casino castle casual cat catalog catch category cattle caught cause caution cave ceiling celery cement census century cereal certain chair chalk champion change chaos chapter charge chase chat cheap check cheese chef cherry chest chicken chief child chimney choice choose chronic chuckle chunk churn cigar cinnamon circle citizen city civil claim clap clarify claw clay clean clerk clever click client cliff climb clinic clip clock clog close cloth cloud clown club clump cluster clutch coach coast coconut code coffee coil coin collect color column combine come comfort comic common company concert conduct confirm congress connect consider control convince cook cool copper copy coral core corn correct cost cotton couch country couple course cousin cover coyote crack cradle craft cram crane crash crater crawl crazy cream credit creek crew cricket crime crisp critic crop cross crouch crowd crucial cruel cruise crumble crunch crush cry crystal cube culture cup cupboard curious current curtain curve cushion custom cute cycle dad damage damp dance danger daring dash daughter dawn day deal debate debris decade december decide decline decorate decrease deer defense define defy degree delay deliver demand demise denial dentist deny depart depend deposit depth deputy derive describe desert design desk despair destroy detail detect develop device devote diagram dial diamond diary dice diesel diet differ digital dignity dilemma dinner dinosaur direct dirt disagree discover disease dish dismiss disorder display distance divert divide divorce dizzy doctor document dog doll dolphin domain donate donkey donor door dose

double dove draft dragon drama drastic draw dream dress drift drill drink drip drive drop drum dry duck dumb dune during dust dutch duty dwarf dynamic eager eagle early earn earth easily east easy echo ecology economy edge edit educate effort egg eight either elbow elder electric elegant element elephant elevator elite else embark embody embrace emerge emotion employ empower empty enable enact end endless endorse enemy energy enforce engage engine enhance enjoy enlist enough enrich enroll ensure enter entire entry envelope episode equal equip era erase erode erosion error erupt escape essay essence estate eternal ethics evidence evil evoke evolve exact example excess exchange excite exclude excuse execute exercise exhaust exhibit exile exist exit exotic expand expect expire explain expose express extend extra eye eyebrow fabric face faculty fade faint faith fall false fame family famous fan fancy fantasy farm fashion fat fatal father fatigue fault favorite feature february federal fee feed feel female fence festival fetch fever few fiber fiction field figure file film filter final find fine finger finish fire firm first fiscal fish fit fitness fix flag flame flash flat flavor flee flight flip float flock floor flower fluid flush fly foam focus fog foil fold follow food foot force forest forget fork fortune forum forward fossil foster found fox fragile frame frequent fresh friend fringe frog front frost frown frozen fruit fuel fun funny furnace fury future gadget gain galaxy gallery game gap garage garbage garden garlic garment gas gasp gate gather gauge gaze general genius genre gentle genuine gesture ghost giant gift giggle ginger giraffe girl give glad glance glare glass glide glimpse globe gloom glory glove glow glue goat goddess gold good goose gorilla gospel gossip govern gown grab grace grain grant grape grass gravity great green grid grief grit grocery group grow grunt guard guess guide guilt guitar gun gym habit hair half hammer hamster hand happy harbor hard harsh harvest hat have hawk hazard head health heart heavy hedgehog height hello helmet help hen hero hidden high hill hint hip hire history hobby hockey hold hole holiday hollow home honey hood hope horn horror horse hospi-

tal host hotel hour hover hub huge human humble humor hundred hungry hunt hurdle hurry hurt husband hybrid ice icon idea identify idle ignore ill illegal illness image imitate immense immune impact impose improve impulse inch include income increase index indicate indoor industry infant inflict inform inhale inherit initial inject injury inmate inner innocent input inquiry insane insect inside inspire install intact interest into invest invite involve iron island isolate issue item ivory jacket jaguar jar jazz jealous jeans jelly jewel job join joke journey joy judge juice jump jungle junior junk just kangaroo keen keep ketchup key kick kid kidney kind kingdom kiss kit kitchen kite kitten kiwi knee knife knock know lab label labor ladder lady lake lamp language laptop large later latin laugh laundry lava law lawn lawsuit layer lazy leader leaf learn leave lecture left leg legal legend leisure lemon lend length lens leopard lesson letter level liar liberty library license life lift light like limb limit link lion liquid list little live lizard load loan lobster local lock logic lonely long loop lottery loud lounge love loyal lucky luggage lumber lunar lunch luxury lyrics machine mad magic magnet maid mail main major make mammal man manage mandate mango mansion manual maple marble march margin marine market marriage mask mass master match material math matrix matter maximum maze meadow mean measure meat mechanic medal media melody melt member memory mention menu mercy merge merit merry mesh message metal method middle midnight milk million mimic mind minimum minor minute miracle mirror misery miss mistake mix mixed mixture mobile model modify mom moment monitor monkey monster month moon moral more morning mosquito mother motion motor mountain mouse move movie much muffin mule multiply muscle museum mushroom music must mutual myself mystery myth naive name napkin narrow nasty nation nature near neck need negative neglect neither nephew nerve nest net network neutral never news next nice night noble noise nominee noodle normal north nose notable note nothing notice novel now nuclear number nurse nut

oak obey object oblige obscure observe obtain obvious occur ocean october odor off offer office often oil okay old olive olympic omit once one onion online only open opera opinion oppose option orange orbit orchard order ordinary organ orient original orphan ostrich other outdoor outer output outside oval oven over own owner oxygen oyster ozone pact paddle page pair palace palm panda panel panic panther paper parade parent park parrot party pass patch path patient patrol pattern pause pave payment peace peanut pear peasant pelican pen penalty pencil people pepper perfect permit person pet phone photo phrase physical piano picnic picture piece pig pigeon pill pilot pink pioneer pipe pistol pitch pizza place planet plastic plate play please pledge pluck plug plunge poem poet point polar pole police pond pony pool popular portion position possible post potato pottery poverty powder power practice praise predict prefer prepare present pretty prevent price pride primary print priority prison private prize problem process produce profit program project promote proof property prosper protect proud provide public pudding pull pulp pulse pumpkin punch pupil puppy purchase purity purpose purse push put puzzle pyramid quality quantum quarter question quick quit quiz quote rabbit raccoon race rack radar radio rail rain raise rally ramp ranch random range rapid rare rate rather raven raw razor ready real reason rebel rebuild recall receive recipe record recycle reduce reflect reform refuse region regret regular reject relax release relief rely remain remember remind remove render renew rent reopen repair repeat replace report require rescue resemble resist resource response result retire retreat return reunion reveal review reward rhythm rib ribbon rice rich ride ridge rifle right rigid ring riot ripple risk ritual rival river road roast robot robust rocket romance roof rookie room rose rotate rough round route royal rubber rude rug rule run runway rural sad saddle sadness safe sail salad salmon salon salt salute same sample sand satisfy satoshi sauce sausage save say scale scan scare scatter scene scheme school science scissors scorpion scout scrap screen script scrub sea search

season seat second secret section security seed seek segment select sell seminar senior sense sentence series service session settle setup seven shadow shaft shallow share shed shell sheriff shield shift shine ship shiver shock shoe shoot shop short shoulder shove shrimp shrug shuffle shy sibling sick side siege sight sign silent silk silly silver similar simple since sing siren sister situate six size skate sketch ski skill skin skirt skull slab slam sleep slender slice slide slight slim slogan slot slow slush small smart smile smoke smooth snack snake snap sniff snow soap soccer social sock soda soft solar soldier solid solution solve someone song soon sorry sort soul sound soup source south space spare spatial spawn speak special speed spell spend sphere spice spider spike spin spirit split spoil sponsor spoon sport spot spray spread spring spy square squeeze squirrel stable stadium staff stage stairs stamp stand start state stay steak steel stem step stereo stick still sting stock stomach stone stool story stove strategy street strike strong struggle student stuff stumble style subject submit subway success such sudden suffer sugar suggest suit summer sun sunny sunset super supply supreme sure surface surge surprise surround survey suspect sustain swallow swamp swap swarm swear sweet swift swim swing switch sword symbol symptom syrup system table tackle tag tail talent talk tank tape target task taste tattoo taxi teach team tell ten tenant tennis tent term test text thank that theme then theory there they thing this thought three thrive throw thumb thunder ticket tide tiger tilt timber time tiny tip tired tissue title toast tobacco today toddler toe together toilet token tomato tomorrow tone tongue tonight tool tooth top topic topple torch tornado tortoise toss total tourist toward tower town toy track trade traffic tragic train transfer trap trash travel tray treat tree trend trial tribe trick trigger trim trip trophy trouble truck true truly trumpet trust truth try tube tuition tumble tuna tunnel turkey turn turtle twelve twenty twice twin twist two type typical ugly umbrella unable unaware uncle uncover under undo unfair unfold unhappy uniform unique unit universe unknown

unlock until unusual unveil update upgrade uphold upon upper upset urban urge usage use used useful useless usual utility vacant vacuum vague valid valley valve van vanish vapor various vast vault vehicle velvet vendor venture venue verb verify version very vessel veteran viable vibrant vicious victory video view village vintage violin virtual virus visa visit visual vital vivid vocal voice void volcano volume vote voyage wage wagon wait walk wall walnut want warfare warm warrior wash wasp waste water wave way wealth weapon wear weasel weather web wedding weekend weird welcome west wet whale what wheat wheel when where whip whisper wide width wife wild will win window wine wing wink winner winter wire wisdom wise wish witness wolf woman wonder wood wool word work world worry worth wrap wreck wrestle wrist write wrong yard year yellow you young youth zebra zero zone zoo

# B Test vectors For Mapping Between Mnemonics and Entropy-Seeds

The test vectors include Entropy-Seeds (using hexadecimal) and corresponding mnemonics.

- Entropy-Seed:
  00000000000000000000000000000000000000000000000000000000000000000000000000000000
  Mnemonic:
  abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon art

- Entropy-Seed:
  7f7f7f

Mnemonic:

legal winner thank year wave sausage worth useful legal winner thank year wave sausage worth useful legal winner thank year wave sausage worth title

- Entropy-Seed:
  8080808080808080808080808080808080808080808080808080808080808080
  Mnemonic:
  letter advice cage absurd amount doctor acoustic avoid letter advice cage absurd amount doctor acoustic avoid letter advice cage absurd amount doctor acoustic bless

- Entropy-Seed:
  ffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff
  Mnemonic:
  zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo vote

- Entropy-Seed:
  68a79eaca2324873eacc50cb9c6eca8cc68ea5d936f98787c60c7ebc74e6ce7c
  Mnemonic:
  hamster diagram private dutch cause delay private meat slide toddler razor book happy fancy gospel tennis maple dilemma loan word shrug inflict delay length

- Entropy-Seed:
  9f6a2878b2520799a44ef18bc7df394e7061a224d2c33cd015b157d746869863
  Mnemonic:
  panda eyebrow bullet gorilla call smoke muffin taste mesh discover soft ostrich alcohol speed nation flash devote level hobby quick inner drive ghost inside

- Entropy-Seed:
  066dca1a2bb7e8a1db2832148ce9933eea0f3ac9548d793112d9a95c9407efad
  Mnemonic:
  all hour make first leader extend hole alien behind guard gospel lava path output census museum junior mass reopen famous sing advance salt reform

- Entropy-Seed:
  f585c11aec520db57dd353c69554b21a89b20fb0650966fa0a9d6f74fd989d8f
  Mnemonic:
  void come effort suffer camp survey warrior heavy shoot primary clutch crush open amazing screen patrol group space point ten exist slush involve unfold

# C Revisit: Test vectors For Mapping Between Mnemonics and Entropy-Seeds

The test vectors include Entropy-Seeds (using hexadecimal) and corresponding mnemonics.

- Entropy-Seed (128-bit):
  00000000000000000000000000000000
  Mnemonic: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon about

- Entropy-Seed (128-bit):
  7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f
  Mnemonic: legal winner thank year wave sausage worth useful legal winner thank yellow

- Entropy-Seed (128-bit):

80808080808080808080808080808080

Mnemonic: letter advice cage absurd amount doctor acoustic avoid letter advice cage above

- Entropy-Seed (128-bit):
  ffffffffffffffffffffffffffffffff

  Mnemonic: zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo wrong

- Entropy-Seed (128-bit):
  9e885d952ad362caeb4efe34a8e91bd2

  Mnemonic: ozone drill grab fiber curtain grace pudding thank cruise elder eight picnic

- Entropy-Seed (128-bit):
  c0ba5a8e914111210f2bd131f3d5e08d

  Mnemonic: scheme spot photo card baby mountain device kick cradle pact join borrow

- Entropy-Seed (128-bit):
  23db8160a31d3e0dca3688ed941adbf3

  Mnemonic: cat swing flag economy stadium alone churn speed unique patch report train

- Entropy-Seed (128-bit):
  f30f8c1da665478f49b001d94c5fc452

  Mnemonic: vessel ladder alter error federal sibling chat ability sun glass valve picture

- Entropy-Seed (160-bit):
  0000000000000000000000000000000000000000

Mnemonic: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon address

- Entropy-Seed (160-bit):
  7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f
  Mnemonic: legal winner thank year wave sausage worth useful legal winner thank year wave sausage wise

- Entropy-Seed (160-bit):
  8080808080808080808080808080808080808080
  Mnemonic: letter advice cage absurd amount doctor acoustic avoid letter advice cage absurd amount doctor accident

- Entropy-Seed (160-bit):
  ffffffffffffffffffffffffffffffffffffffff
  Mnemonic: zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo wrist

- Entropy-Seed (160-bit):
  21d91855df6dc86effb9d7c2faac126d106d2fd4
  Mnemonic: canyon silver betray salute symptom danger year depend seed step afraid surface almost obey eye

- Entropy-Seed (160-bit):
  29aa43aa8f6ca8d63629387014e60db516f0ba07
  Mnemonic: claim faith turkey burst skin help suffer check hybrid poet lock health humor fringe buffalo

- Entropy-Seed (160-bit):
  22b733aa14787ce84f59473a79a854d85a3c2383
  Mnemonic: cargo ridge tunnel cigar march injury dice cigar denial snap benefit radar phrase balcony auto

- Entropy-Seed (160-bit):
  7b9e5ae9514db243879748dc12f5738540b2e268
  Mnemonic: knee version ripple penalty swamp canyon bundle trouble swing nut friend appear arctic illness physical

- Entropy-Seed (192-bit):
  000000000000000000000000000000000000000000000000
  Mnemonic: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon agent

- Entropy-Seed (192-bit):
  7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f
  Mnemonic: legal winner thank year wave sausage worth useful legal winner thank year wave sausage worth useful legal will

- Entropy-Seed (192-bit):
  808080808080808080808080808080808080808080808080
  Mnemonic: letter advice cage absurd amount doctor acoustic avoid letter advice cage absurd amount doctor acoustic avoid letter always

- Entropy-Seed (192-bit):
  ffffffffffffffffffffffffffffffffffffffffffffffff
  Mnemonic: zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo when

- Entropy-Seed (192-bit):
  6610b25967cdcca9d59875f5cb50b0ea75433311869e930b
  Mnemonic: gravity machine north sort system female filter attitude volume fold club stay feature office ecology stable narrow fog

- Entropy-Seed (192-bit):
  6d9be1ee6ebd27a258115aad99b7317b9c8d28b6d76431c3
  Mnemonic: horn tenant knee talent sponsor spell gate clip pulse soap slush warm silver nephew swap uncle crack brave

- Entropy-Seed (192-bit):
  8197a4a47f0425faeaa69deebc05ca29c0a5b5cc76ceacc0
  Mnemonic: light rule cinnamon wrap drastic word pride squirrel upgrade then income fatal apart sustain crack supply proud access

- Entropy-Seed (192-bit):
  c10ec20dc3cd9f652c7fac2f1230f7a3c828389a14392f05
  Mnemonic: scissors invite lock maple supreme raw rapid void congress muscle digital elegant little brisk hair mango congress clump

- Entropy-Seed (224-bit):
  00000000000000000000000000000000000000000000000000000000
  Mnemonic: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon admit

- Entropy-Seed (224-bit):
  7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f
  Mnemonic: legal winner thank year wave sausage worth useful legal winner thank year wave sausage worth useful legal winner thank year viable

- Entropy-Seed (224-bit):
  80808080808080808080808080808080808080808080808080808080
  Mnemonic: letter advice cage absurd amount doctor acoustic avoid letter advice cage absurd amount doctor acoustic avoid letter advice cage absurd

apart

- Entropy-Seed (224-bit):
  ffffffffffffffffffffffffffffffffffffffffffffffff
  Mnemonic: zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo veteran

- Entropy-Seed (224-bit):
  c74a8dba069392e820d7e989a7e0248f18b831ae4204e6beb1f59ae9
  Mnemonic: shrug federal hub artwork decrease inject local wood meadow disease ankle bullet merit boat ribbon dog trade void latin cupboard ordinary

- Entropy-Seed (224-bit):
  a9e185b399824423cf1f38aa95638a204e66cef5a805f59d6d053f14
  Mnemonic: prepare around hollow credit cattle ball detect train pride prison tip dog toy guess stove absent void twin donor lawsuit eagle

- Entropy-Seed (224-bit):
  ab20e012dd314be93467e5a2aee00a2f5acea30ea7dfe505cb15f1a2
  Mnemonic: private alpha again risk believe virus spice witness pencil jacket action gadget provide face dentist winner topic comic shell today cinnamon

- Entropy-Seed (224-bit):
  c4b96fe3a9b80d3390742fbcb5bdf1ea7fb6818c680b665ab88632ff
  Mnemonic: seven slice wedding fat liar oil dove magnet royal pupil ladder stay window dolphin coyote accident green still canoe grass vacant

- Entropy-Seed (256-bit):
  0000000000000000000000000000000000000000000000000000000000000000
  Mnemonic: abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon abandon

abandon abandon abandon abandon abandon abandon abandon abandon art

- Entropy-Seed (256-bit):
  7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f
  Mnemonic: legal winner thank year wave sausage worth useful legal winner thank year wave sausage worth useful legal winner thank year wave sausage worth title

- Entropy-Seed (256-bit):
  8080808080808080808080808080808080808080808080808080808080808080
  Mnemonic: letter advice cage absurd amount doctor acoustic avoid letter advice cage absurd amount doctor acoustic avoid letter advice cage absurd amount doctor acoustic bless

- Entropy-Seed (256-bit):
  ffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff
  Mnemonic: zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo zoo vote

- Entropy-Seed (256-bit):
  68a79eaca2324873eacc50cb9c6eca8cc68ea5d936f98787c60c7ebc74e6ce7c
  Mnemonic: hamster diagram private dutch cause delay private meat slide toddler razor book happy fancy gospel tennis maple dilemma loan word shrug inflict delay length

- Entropy-Seed (256-bit):
  9f6a2878b2520799a44ef18bc7df394e7061a224d2c33cd015b157d746869863
  Mnemonic: panda eyebrow bullet gorilla call smoke muffin taste mesh discover soft ostrich alcohol speed nation flash devote level hobby quick inner

drive ghost inside

- Entropy-Seed (256-bit):
  066dca1a2bb7e8a1db2832148ce9933eea0f3ac9548d793112d9a95c9407efad
  Mnemonic: all hour make first leader extend hole alien behind guard gospel lava path output census museum junior mass reopen famous sing advance salt reform

- Entropy-Seed (256-bit):
  f585c11aec520db57dd353c69554b21a89b20fb0650966fa0a9d6f74fd989d8f
  Mnemonic: void come effort suffer camp survey warrior heavy shoot primary clutch crush open amazing screen patrol group space point ten exist slush involve unfold

# D Test vectors For Derivation of Master-Seeds and Account-Root-Seeds

The test vectors include Entropy-Seeds shown in Appendix B and corresponding Master-Seeds. In addition, Public Rands for some randomly chosen Sequence Numbers are given. All item are using hexadecimal.

- Entropy-Seed:
  0000000000000000000000000000000000000000000000000000000000000000
  Master-Seed:
  d4fde696ab58de7b5097a1ec017e59d1f440342c1e278ea092d2c88cec6cc147
  coinSpKeyRootSeed:
  56c658bad0035e16677d561b22ffb56e194f7160e40eb37466a8afbde5cb7bd7
  6d4061b38a67f32b63b6e03f2b6b3e49a55671170990ee01be672bd4e0356632
  coinSnKeyRootSeed:

bddd5c6a3049d96124666f009c0fb3af2b695fd28dd567b5eb25130b24788d20
21edeae091cc10ae5154878dc94098c22e60f4a663efaa98c6916d055f7802ca
coinDetectorRootKey:
a2f568513bcd1610d4cca7f9770b84f7815c16a011a4fd27387f5e9067191f4f
6b823290c0d8a394a412b5d068d3385e5a71cefb541c529e74c7215058f0ba0d
coinVKRootSeed:
e3a7966dd6e343509b682be724be4035ff8ac480775f4dd72559aa63f6124e56
9d5341321c43a68eec7c1cb9232cae7bd643da63c9306cdba7e3415315bd48b2
publicRandRootSeed:
48d41357356d035064d5abb63b166413d362ef61f43bf333fffc5351dbbe977f
b383e743727f0360b5e16cd1e5fe7fbda816d20e15b869e1bc90acee79962673

Public Rand for Sequence Number 887699001:
d832f4df91d59bf2c26c3dc75975c1a53b484089726bd4a82f10b46cbb6f56af
094374b37d0953350af428153e99873a10265269db2b9ce4b30b0c292f7a3136
Public Rand for Sequence Number 2172391158:
dfa3ecb99d8c6ba43cd0fa4d778f910333125ab47833dc83279785dc87ee634f
dd0562d6aa43004f6db3f94efcbdae4ace5f1b4b738de815c299e68aacf06928
Public Rand for Sequence Number 2474215398:
10d5530ffa173f24bdc2a2970e42403d55a676d2d29649f56f9e47458ff04681
c1cbc08d6f15bb6fa34dea9d273b50e38da3047c8652dabe65d3f050e8ec6224

coinValueKeyRootSeedAut:
7c7ac669e03127f1c040ec46d92d7d273f0ed2acd218f33b1dc56508934210e3
ed06b7147dce709ac357ae90712d7b5e3cbcf1dc6bf7e4809a05f03d5b6787ea

- Entropy-Seed:
  7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f7f
  Master-Seed:
  23ea3cb377aacebca6bc70e8b702b9098e7edd3f9fba5bc0ac03ff193797e83e
  coinSpKeyRootSeed:
  c80dc924e3ea9c81c0284e9361d63ab3f1faedfe83a5afbffbfd42fe1a0244c9
  adacf693290ee764bf12b99e1c07501919096fdd6778e009bdccee203331385e
  coinSnKeyRootSeed:
  07d0cc2b43f66ef4c082cb633c279550270075f06242c78cfe7ba10a8d48e0df
  f8eaf1683868af72f5ff2320b8df05d4a486355ff1319a5d359cf72fe45fa4dd
  coinDetectorRootKey:
  32db13d53978947e0295e5ce056daca0db2d14266a83eee1687894a6239b7cde
  6e2414d966599e1f2e8253a223714eacc7f0cbd17c144264905457873acf2989
  coinVKRootSeed:
  7d32f4fcfac883ac7ef93e9fb1ebeaf3ea51086551245e49f7b4f28c052bd661
  822d95387c56ba1857661d59ad66b30fb4a8c434a9a7bfbb673bc76b64e681db
  publicRandRootSeed:
  2d8dd5f03ae6de9a9852a8ebaa3407819437e426df170e3d1ed23937e8f601ab
  7a0efa8ddf5c75b58bf99b11ccea6e66242afe4d49f4b46795bee592c480096b

  Public Rand for Sequence Number 202015495:
  3276ef36fe147d736fef8cb4008de0cf58ef18dedb298d3c70577be74163e7f1
  bb3c2948fda6d77b1ed31124717470b790ad87e7180dd41f9d6239f7fdf9f17a
  Public Rand for Sequence Number 4184040278:
  93d104339ba3a1fe765c7150a0c353dbbdd16feb44a564278d5392f191bdf62c
  1250a193e3ea4fecb74effe5b4c4bfc780058fc7de65db9deab64873fbd91388
  Public Rand for Sequence Number 3136240371:

96744b584bc009bc451720f16ded480be573095bbc9a2af5685d9455fcd46c12
71e1314f3bfafae9a0abfbc50b1114a0b812814750632e09c3ec6a1bcf0cd789

coinValueKeyRootSeedAut:
98d10c98e5cc99c014539b97e93526d60d52f142ed553347c42c77388d886b7d
21d9097756276562a895b142d633841883528c0b3ad3e56fe840bfa5ae390477

- Entropy-Seed:
  8080808080808080808080808080808080808080808080808080808080808080
  Master-Seed:
  967ae5d533b0560b01c713f08e6e8fc5cac7e106e029c8a6ad835e68ae94db5c
  coinSpKeyRootSeed:
  9c45a797fc2d98fa6d259d2387565dde886d1a7fd316b05f586ec60b04c6062c
  a06efc1bcc56c620c11b087ec2068a64cdc9e8fe9fd96c1fbd6462b7c0e463dc
  coinSnKeyRootSeed:
  6b9caf9485bc375c5d810a2cc07774b688a736ff5ce8023ed22f7467ab51f43c
  106eaf975fdcd1b2c0eb7afe6df5c16242dab027f14cdbe3b057e05e58c448b3
  coinDetectorRootKey:
  57e389fafad8cdd8d496fc6fde89e5ebb1cb4dc96fcb1cdbd58e757579603b66
  d13daa40e9ba315bde8aefc1c4cfcae7e2bd45c98eb6289ab306b30fdc7d770d
  coinVKRootSeed:
  d8e0a5709e36a1b2fdb18b3c592859f23ca9743911396231be1b97ca6f85b8b1
  24b30013773b5b0e44ba65dd566a81f6697e52e2897b0375f6761519e19da4fa
  publicRandRootSeed:
  6af846a26566e6dcd5e55dc5549fda24a1afd9e420becbc383228db6cbb18c21
  436e0b0b9e7c331359e84efddc5fa06bebcc7a43e0f454becf8962cb5d9d4631

Public Rand for Sequence Number 2400773142:
1823c3c895c419aa32b993d788e87a722358d24dd57e05f0ab0c749e39a5be48
e0075fdc07038eb9e1adbc274dd7cfa3d313ec97fcdefaff56aa9770d6046fd6
Public Rand for Sequence Number 94650305:
d2e984e12d818ba294d0571722d51d300ce32549016f0bc93cb1850b811d9b61
9f34bf3aa909c7d4a65253202fe11148cefd57a6b42dcb93891d94123e1dc474
Public Rand for Sequence Number 3668972267:
bc160f59c1885986b185540d8628485645ece76952bef6afde49ae7a5c1b5d89
f8e3118ecf0d4f3d46ddae7cf95b88c9a9b49fe170a0696a6de9027782bde90e

coinValueKeyRootSeedAut:
81a723c11135ee7c4308ba47149c419ebb069110b44f986b656f648429e7d2fd
d2f3e3cc7855b504bf6d52528e2e5014c9c15375d23172f12eac9ce8a92a39f8

- Entropy-Seed:
ffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff
Master-Seed:
4abcc8067fefc753596a079a6b2c81bbff92f65c0bae6369f7572f8cfe881161
coinSpKeyRootSeed:
a46e06fa6d94789a147f22d4862e39b40a0d2cc05ca26738705cfe2d399d3c87
b71f8d99dfc3900281a3b569a5ed2f54f0ee4d0d807593599bd2bedc170d4b7c
coinSnKeyRootSeed:
ae609c4c2ee8d33395452d0e03cb6be58a939ca6f28095235622cf542a51f407
b0a2f31a8bdff957960518be5a1e2b8c0be53d98b75947525754a6820211ce47
coinDetectorRootKey:
7c32ecbc44e49533ce29946314848d60b5e432080a3de6db3b0abaab983a4a55
d40744b86a10817347e89588ab622bc9a8369b3958d7f42b49b263992238cc38

coinVKRootSeed:
110851bc774a02d79b8df4896fd4e497befc1c7bb8d447d282fe94bb13574fb4
ad68bca6337dd8575e907d4a94ef4dc2d180c9f4e99d567b19ca2d9e4459a214
publicRandRootSeed:
5b14a23daa5936f5828e5a7117dc1fd76c4aab6a23a57de8419ee52ca253021a
aa6a4ccd6468e57ebc5ffb2e672171cd0216febd76fe1e8ab07fe16ca2775d71

Public Rand for Sequence Number 4151069018:
577ca99203857d5b233abafbee3327e7ea2d4af5d0d64014a4ac047a8f2d911c
698bbee62e8d19e5738f615730b594e86f0098f02b29ef2d88ae3a9e92253a7d
Public Rand for Sequence Number 810230981:
1ed8f5a3bbdce1201782376807364ff46424bcebd43a01f6b8074c97c4c4cc57
6b497a8931ae874d08ae5aa04b4420d771e79a2a97a2636ee20d880424df6b86
Public Rand for Sequence Number 1205304842:
ef6df803d5b2ab20050752cdc59f7c4e16f11172e86782092dc15413484026b7
aed522747178ddcc5b1b57f3067636b9392a41117a2975216d7bd34448eb82c5

coinValueKeyRootSeedAut:
59b9400f1448564bccaa1a4c679810844fcaeef76a3e6fc6b75ba39e18485025
789b917766cd589e3173e950a9b0bc4db78b9fd8a150bb46be1ca795d4e4e415

- Entropy-Seed:
  68a79eaca2324873eacc50cb9c6eca8cc68ea5d936f98787c60c7ebc74e6ce7c
  Master-Seed:
  ec243d0407f4731085ecd7bef047db836b675e8622c937454c3313a8129d6733
  coinSpKeyRootSeed:

b3b6835aa6dcbe526ff030bca5e71a825a23ff1605902f39e392971df493c30b
2dd3c74082a97351bc3b0a152e1563f724ddbbf2adb8c2906e00c77d235b5515
coinSnKeyRootSeed:
becde156316d7470eacdc12d01452fc56aa27bb6f1a7ace1ba1a26da3752aaba
f819d97609513fc11e4f8e34a631e062d6467626d47917e0438fdf3c3eb59ccb
coinDetectorRootKey:
ed54c3155e21d98f316fb521c92b09679e763e9b80695d62ab76cc17c57ece57
86b54d83494f0e81967ff37c772f7dbcb830069665ad2fc95b83acfe6f9d3ea7
coinVKRootSeed:
f4b510fce8ad25bb3e8e2e3a371a813cd8557d1930284c1b54b301b481965232
e4e36a91c19c622eba1625926d4df7654fbe51036d0af31bc6a93e1fd9f09932
publicRandRootSeed:
1a8efaa57603373aa226bd25a734b355570f777dd3a50b113cdea7ce3902537c
a80682f918a4c26796b4f3a2c64ceeab877d091adac6b1083237fdef19a81d5b

Public Rand for Sequence Number 1386746106:
d6ed41486c223a9bc0877b1465d2ea412ecd7dc5d4aa6e7f5caaf65c1265d7f9
30e55c7c626715d7a9c614b00be192fbbae3293f8440e553f079ab35b24510db
Public Rand for Sequence Number 757380707:
2245d0ecf4e2ddffbeb80fbef9b507e35561acce80451c85e548eb431123ecd7
032d34b182a381f5356db1dd5e92d2c0b40a36cb3e3a98b6ecdf81ae65f6e1e5
Public Rand for Sequence Number 1590126875:
f069a5a75a4563e6e190e8f8b1443341ce3272b68cff99ff384ac47478541dc5
b061fcc78fd750fa3beea3a6e86530691cada8af012d1c12b34344bdab0caaeb

coinValueKeyRootSeedAut:
a230714e5df98c856dedf7a387ac486b5ffa8fabf562d3563fd45b27ef16322c

d15e2b3726300215b4c597b3bed385060a0a280c009cd5bbbbefbc041e0f56c1

- Entropy-Seed:
  9f6a2878b2520799a44ef18bc7df394e7061a224d2c33cd015b157d746869863
  Master-Seed:
  cd33cfeaec5a48f52f485fea5e9637889609aa64cab852d80aa258e989071c60
  coinSpKeyRootSeed:
  a02c93c369274f3d866e98cf6980dfbac25587cbbb4c8b5f45c3f62e2e94b214
  913dcbe832a2908511b9d1373feb6ef1df42d147bc11a9e199d3d51638ca771f
  coinSnKeyRootSeed:
  babca8531d788cbec214e9a3180301479b2df429705aeb8e454630c5d54b0fcc
  dc831f5a1c12bc3c8d92ac3ad37e0c195a7ea3041ff39d3bb855f02eadfb57c7
  coinDetectorRootKey:
  b2c572f0108cc12627d17f8b93e7dd86f0545ee568309c2c93ecf4d514b2501b
  b1e596e2bd4005d3068fe96f430b5564cd393aabfd792c40554bb9a09c8e1406
  coinVKRootSeed:
  e02c06fbe97da236dcfb098b256f4edf4727f0b19cf5f0afe7f4ed9ea524dab7
  1adbfd0536267e7c8ebe0909d4820de92f5c37d3bfca0aeb1c34a210d9be189d
  publicRandRootSeed:
  88087c0f9482aed16a34017404c45541055be1624993b727d7296be581fa4d11
  05044b680ab2f00f8dfdb8f6f7712580082e6196e09179b27b529e9b1310d8fb

  Public Rand for Sequence Number 1349967783:
  1ab49a08f7dc26c37d10e78d9afdd81c08601391b31a81add08ffdf937444953
  1e16876deeaac3d1d896af37f26d0e8b4845205e11dc508c2c4789f0c1150aa2
  Public Rand for Sequence Number 2895057325:
  3eea3875fb600d6adf5134bb61ee794cd250441315dab228fef0ed070dd172e2

9382e62c1e09d1dd4f6b60113effed91d3a9c98c606d05307aeb644fc2c62402
Public Rand for Sequence Number 335187303:
627796c337060b19ec1c7178b4630f08062c6259318ad20b04d7f3308dff00b9
416dfc75652cf3e63d63bec1bdf99599edc22c78003a885d1c6c42a94348d5bf

coinValueKeyRootSeedAut:
3acda1d4b30360b2a200300238638f12a4e037eb2ab86e48b290afbd02bab60b
d0b5e4e7bffa8f24359acd5c5ce53ec1b3328ed3f83ab6a252cab464228ca50f

- Entropy-Seed:
066dca1a2bb7e8a1db2832148ce9933eea0f3ac9548d793112d9a95c9407efad
Master-Seed:
7242f5fd0c0b8df7429d60a7a757687eb486e4f45f50f675a9d23fa034607f3a
coinSpKeyRootSeed:
c63f57958e0b02af215837e7a9f7da3f6830ca849e9cc586a1016f59d518ff2d
b832d9b3a70fa8f46f91d63af9ecf4681a639683c8b178a3e4cc4d0ad2e2fdab
coinSnKeyRootSeed:
d1d62e7f9bee336a61e360785bf1fca64dc9b90602b99abeaff44f48c3dfdc74
14d2a23dca36c929cbf093b27e63d6797797ffb8ff2c3862c90d3dbf4fa9a924
coinDetectorRootKey:
ae85d5eafd21843dc6f7518861982107fa63a2a95ab4eb09d0c7b50e5d4f3f9a
afed0f23b2a1958c27f1f5d0d27e23f1716a9093abe2430de1f028f51a397b73
coinVKRootSeed:
dff3840573a631233b893c59c33518324f84c6da4e13cc19dff3aaca66285b09
14dee95e6e8b6f2d1cba3d8ac46beebec09f7b7fadfd1a358cdd5bc26aa84a8d
publicRandRootSeed:
70f94f5a37f48e64c8ccb91779d131c6fbd25f44e3a463a196f21de7502f2c5f

5aa9461361af03d1fc560df9ccb96c1afaf1758e9112a01d555cdf42ead3b7de

Public Rand for Sequence Number 3037234089:
4fee704d8d687a026cfd98d1a19dc0cc1d7298b350f9115a722e4388d204e903
6c8b698e2870ca8e04d5fc7716764ad16278022179d05bca86e8a5fdb0941b2c
Public Rand for Sequence Number 3952040230:
77a47010286d7b308e0ec22dfac6768f3a44a430ca54e11fcf72be6cf3379b4f
a30f16a246e404166d2b13c02dc16e9300d84baf3986bd10d2eae731d1a43d52
Public Rand for Sequence Number 997927870:
d58a8e1f7e19a4f5ed092636be3477284a24d7b9d4d928776512123d29194579
6f3e286f097bac617faf946d5d55286eb7c670a26f382b0a6e1e27092f36a291

coinValueKeyRootSeedAut:
4b4c30616d2a0b03fbae33ac9e172d4150a236b4ac416b7e11518b4b5bebd600
dd3fbf9fe01759e1e56371f2dc9e14becc7058b58494fa1be57affa6c89af73c

- Entropy-Seed:
  f585c11aec520db57dd353c69554b21a89b20fb0650966fa0a9d6f74fd989d8f
  Master-Seed:
  e82bbf95ad7518c82df7165f575d07634edb48d9eb63e444a104e625dbca0e16
  coinSpKeyRootSeed:
  3f185e15f879378bc7df9c604fe966024f2aa46101eb9aecf873a556126ad924
  c9d8f1bce9a28238989404924cf1ee571b8f06015843010a38ed1fca6d9263e2
  coinSnKeyRootSeed:
  0ea3fc5da6fff18dfe3dfc9a87a394ac7543fa2f0cf26e5d8b7e2ec7bd3f7ac9
  6d063217f60faa92d1de41b2504684219852de154b81ab3a507ba02562957591

coinDetectorRootKey:
a808ef8b153c4f53ad1dc6a64c2c2fea2998cdb41c2abdf1e4659bc07a9369a0
048f4b5fd82385f7dc31780e789e6eed020cff9d0eacddd0faefe6d86e61e4ec
coinVKRootSeed:
8c8f88993d54e24c2d676042df3a3839d406d7d0d5166ca6d05a52d1aad0b04d
c539f9194dcf492afce2ce1711b0c8e92e0660a54698c94c86835219304084c4
publicRandRootSeed:
9f3e551aeafc8304816980d6c1e3fd4c10d24635df07900cf5dddfbe72a1e98d
8308fc4cb7acf827e21136d352b92ba79bf8bdbbca45fc367817f65ec39a0367

Public Rand for Sequence Number 2282586443:
13abd9c51cf91e7ffd7f90daf2c1bf3a1210bba87d75e4cfe09fa285a08c4079
aea3efdcaa83f7a83df7a265d709bd6f817670a4791585e7011b461912479788
Public Rand for Sequence Number 2058736541:
2b51f39a3e6a049a5ab3d856962869580f889287758f32ebe7a5185f620da93f
c4df5e62ce2d0b11295e3ad0e707df306c3c21ab45fe718c431e7c6ef8eb6c73
Public Rand for Sequence Number 3650681307:
b3923c0758fe02869413a515dbc8fd129b2b876f472c8db3ec114455fb21d37d
3d97cf54f3f7540e182a610507e04dce7056bf8b294bb2bd887c22b7c9c742ca

coinValueKeyRootSeedAut:
d5d86a1e486ea52013b2a6967a65b16370b1cee6748ec92a6cd5bf45726e2758
e2cf9f8b7935c825c472ee93a5895ba2ce697d021e84f0110f849604b7cc8b7b